

Smart LPG Gas Detection & Automatic Booking System

Detailed Project Report

An embedded safety and gas-cylinder monitoring system that detects LPG leakage, monitors cylinder weight, provides local alerts, and uses a GSM module to send SMS notifications and an automatic booking request when the gas level becomes low.

1. Abstract

This project presents a microcontroller-based LPG gas monitoring and automatic booking system. An LPG gas sensor continuously monitors for leakage while a load cell with an HX711 amplifier measures the approximate cylinder weight. A 16×2 LCD displays the gas level and measured weight. When a gas leak is detected, the system activates a buzzer and relay and sends an SMS warning to a predefined alert contact through a GSM module. When the measured cylinder weight falls below a configured threshold, the system sends a low-level alert and an SMS booking request to a predefined booking number. The design combines safety monitoring and refill notification in one embedded system.

2. Problem Statement

Conventional domestic LPG usage generally depends on manual observation for detecting leakage and remembering when a cylinder is nearly empty. A gas leak can create a serious safety risk, while a low cylinder level can cause unexpected interruption of cooking. The proposed system aims to provide continuous monitoring, immediate local warning, remote SMS notification, and automatic refill-request messaging based on cylinder weight.

3. Objectives

- Detect LPG leakage using an analog gas sensor.
- Measure LPG cylinder weight using a load cell and HX711 ADC module.
- Display weight and gas-level information on a 16×2 LCD.
- Activate a buzzer and relay during a detected gas-leak condition.
- Send an SMS warning to an alert contact when leakage is detected.
- Detect a low-cylinder condition using a configurable weight threshold.
- Send a low-level SMS alert and an automatic booking request through GSM.
- Provide a simple embedded implementation suitable for domestic LPG monitoring.

4. System Components

Component	Purpose
Microcontroller board	Reads sensors, controls outputs, LCD, and GSM communication.
LPG gas sensor	Provides an analog signal representing the detected gas level.
Load cell	Measures the cylinder load/weight.
HX711 ADC module	Amplifies and digitizes the load-cell signal.
16×2 LCD	Displays measured weight, gas level, and status messages.
SIM800L / compatible GSM module	Sends SMS alerts and booking messages.
Buzzer	Provides a local audible warning during gas leakage.
Relay	Provides switched output during the gas-leak condition.

Component	Purpose
Mode switch	Connected as an input using the internal pull-up configuration.
Power supply & wiring	Provides power and electrical connections for the modules.

5. Pin Configuration Used in the Code

Device / Signal	Pin
LCD RS	D13
LCD EN	D12
LCD D4–D7	D11, D10, D9, D8
HX711 DT / DOUT	A0
HX711 SCK	A1
Gas sensor analog output	A2
Mode switch	A3
Buzzer	D7
Relay	D6
GSM SoftwareSerial RX/TX	D2 / D3

6. Working Principle

Step 1 – Initialization: The microcontroller starts Serial communication, initializes the LCD, GSM serial interface, and HX711 load-cell ADC. The load cell is assigned the calibration factor used in the code. The GSM module is given time to initialize and a test SMS is sent to the alert contact.

Step 2 – Weight measurement: The HX711 is continuously updated and the cylinder weight is read. Negative readings are forced to zero. The measured value is displayed on the LCD and printed to Serial Monitor.

Step 3 – Gas monitoring: The analog gas-sensor reading is acquired from A2. The code scales the raw analog value by dividing it by 10 and displays the resulting gas-level value.

Step 4 – Leakage response: When the calculated gas value reaches 3 or above, a leak condition is declared. The buzzer and relay are activated, an alarm status is shown, and an SMS warning is sent. A flag prevents repeated SMS transmission while the same alarm condition remains active.

Step 5 – Low-cylinder response: When measured weight is 2 or below, the system displays a low-level status, sends a low-gas SMS to the alert contact, and sends a booking-request SMS to the predefined booking number. A second flag prevents repeated booking messages until the weight rises above the threshold.

7. Software Logic

1. Initialize LCD, GSM, Serial Monitor, and HX711.
2. Read cylinder weight.
3. Read analog gas-sensor value and scale it.
4. Display weight and gas level.
5. If gas level $\geq 3 \rightarrow$ leakage alarm + buzzer + relay + SMS.

6. If gas level returns below 3 → clear leakage alarm.
7. If weight ≤ 2 → low-level alert + booking SMS.
8. If weight rises above 2 → clear low-level flag.
9. Repeat the monitoring cycle.

8. GSM Communication

The project uses SoftwareSerial for communication with the GSM module. The code switches the GSM module to SMS text mode using AT+CMGF=1, sends the recipient number using AT+CMGS, transmits the message text, and terminates the message with ASCII character 26 (Ctrl+Z). A helper function also waits for a response from the GSM module and reports a timeout if no response is received.

9. Alert Conditions

Condition	System Response
Gas value ≥ 3	Leak alarm; buzzer ON; relay ON; warning SMS sent.
Gas value < 3 after alarm	Buzzer OFF; relay OFF; alarm flag cleared.
Weight ≤ 2	Low-level status; low-gas SMS; booking SMS sent.
Weight > 2 after low-level condition	Low-level flag cleared.

10. Key Programming Concepts Demonstrated

- Embedded C/C++ style Arduino programming.
- Analog sensor acquisition using analogRead().
- Digital output control using pinMode() and digitalWrite().
- LCD interfacing using the LiquidCrystal library.
- Load-cell interfacing using the HX711_ADC library.
- Serial and SoftwareSerial communication.
- AT-command based GSM SMS communication.
- State flags to avoid repeated alerts.
- Sensor threshold comparison and event-based control.
- Basic GSM response handling.

11. Strengths

- Combines safety monitoring and cylinder-level monitoring.
- Provides both local and remote notification.
- Automatic booking reduces dependence on manual checking.
- Low-level and leakage events are handled independently.
- Simple threshold-based logic is easy to understand and maintain.

12. Limitations and Practical Considerations

- The gas value used by the program is a scaled ADC reading, not a calibrated gas concentration such as ppm.
- The weight threshold and HX711 calibration factor must be calibrated for the actual load cell and cylinder setup.
- The relay output is controlled as part of the alarm response; the exact connected load and relay behavior

must be designed safely.

- The GSM module requires suitable power supply, SIM/network availability, and correct antenna configuration.
- The code currently contains phone numbers and a connection number directly in the source code; these should be replaced with placeholders before publishing publicly.
- The test SMS in `setup()` causes a message to be sent whenever the controller restarts.

13. Future Enhancements

- Use a calibrated LPG sensor and provide a meaningful gas-concentration display.
- Add a proper rechargeable/backup power arrangement.
- Add mobile-app or IoT dashboard monitoring.
- Store historical weight and leakage events.
- Add multiple alert contacts and configurable thresholds.
- Add safer automatic gas-shutoff hardware where appropriate.
- Add fault detection for sensor disconnection and GSM communication failure.
- Provide configuration for changing alert and booking numbers without modifying firmware.

14. Conclusion

The Smart LPG Gas Detection and Automatic Booking System demonstrates a practical embedded-system approach to improving LPG safety and availability. By combining gas sensing, load-cell measurement, LCD feedback, local alarm control, and GSM-based SMS communication, the system can identify two important conditions: possible gas leakage and a low cylinder level. The project also demonstrates useful real-world embedded concepts including sensor interfacing, calibration, threshold-based decision making, actuator control, and GSM AT-command communication.